Auditing PPI Graph Augmentation for Missense Variant Pathogenicity Prediction under Topology, Connectivity and Homology Constraints

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***Abstract*—Protein-protein interaction (PPI) graphs provide a natural source of molecular context for missense variant pathogenicity prediction, but performance gains obtained by propagating protein embeddings over a shared graph can be difficult to interpret. Such gains may reflect protein-topology alignment, train-test connectivity, homology-related split structure, or complementarity to existing variant-effect predictors. We audit these possibilities using a reconstructed ClinVar-STRING dataset of 177,213 filtered, non-conflicting ClinVar missense variants mapped to 14,123 labelled STRING proteins within a 19,622-node STRING protein-association graph used as PPI context. ESM protein language model embeddings are compared with fixed two-hop PPI-GCN propagation, CADD, AlphaMissense and fusion models using validation-selected downstream MLP classifiers. Across three classifier seeds, PPI-GCN increased mean AUC from 0.7835 to 0.7950 and mean AUPRC from 0.6116 to 0.6484. In a retained reference run used for paired graph-ablation analyses, removing direct train-test PPI edges nearly eliminated the AUC gain, protein-clustered bootstrap widened uncertainty across zero, and topology permutation returned performance close to the raw ESM range. Graph fusion also improved CADD +AlphaMissense under the original split, but follow-up controls showed that much of this gain was retained by raw-ESM, edge-ablated and topology-permuted protein representations, indicating limited topology-specific added value in the strong-predictor fusion setting. In the homology-aware stress test, the retained graph-fusion estimate did not exceed the repeated-run architecture-matched CADD+AlphaMissense baseline. Topology-stratified and sample-level transition analyses suggested that graph-associated gains and prediction changes were concentrated in specific network roles and error types. PPI graph augmentation acted as a context-dependent source of network information, with added value shaped by graph connectivity, split design and baseline predictor strength.**

I. Introduction

Missense variant pathogenicity prediction has been advanced by established methods such as CADD, REVEL, MVP, PrimateAI, EVE and AlphaMissense [12], [9], [18], [23], [7], [5], as well as by protein language models such as ESM [14]. These approaches capture sequence, evolutionary and annotation-derived signals. Disease-associated proteins are also embedded in molecular interaction networks, and disease-associated mutations can perturb protein-protein and other macromolecular interactions [1], [16], [19], [3]. PPI graphs provide an additional source of context for missense variant prediction.

Earlier approaches incorporated protein-interaction and other molecular-network features into single-amino-acid variant prioritization, showing that network context can provide information complementary to sequence, structure and annotation features [26], [17]. More recent graph-based variant predictors include graph attention models for missense effects, heterogeneous gene-variant graph approaches, multimodal variant graph methods and disease-specific graph models [28], [4], [27], [8]. These studies show that relational structure can support variant interpretation, but they do not resolve a narrower evaluation question: when protein language model embeddings are propagated over a shared STRING PPI graph before variant-level classification, what exactly does the observed gain mean?

This question matters because a shared-graph setting can mix several effects. Train, validation and test variants may have no overlapping protein identifiers, yet their proteins can remain directly connected in the same PPI graph. Related proteins may also cross partitions unless homology is explicitly considered [15], [25]. Graph propagation may add little once strong variant-effect predictors such as CADD and AlphaMissense are already present. Evaluation must move beyond a single AUC or AUPRC gain and determine whether the gain is consistent with protein-topology correspondence, sensitive to graph connectivity, robust to protein-level uncertainty, stable under homology-aware evaluation and complementary to established predictors.

We frame the work as an audit rather than a new predictor proposal. Using a reconstructed ClinVar-STRING missense dataset, we evaluate raw ESM representations, PPI-GCN propagation, CADD, AlphaMissense and fusion models. The analysis combines variant-level and protein-clustered bootstrap, train-test edge ablation, partial edge-ablation curves, topology permutation, homology-aware splitting, topology-stratified analyses and corrected-sample profiling. It tests the dependence of raw-ESM-to-PPI-GCN gains on protein-topology correspondence and cross-split connectivity, examines how strong variant-effect predictors and homology-aware splitting bound the apparent gain, and localizes graph-associated improvements and errors to specific topology contexts. PPI graph augmentation can improve benchmark scores, but the gain depends on the baseline, split design and graph-control setting, and its added value is bounded by homology-aware evaluation and strong external predictors.

II. Materials and Methods

A. Dataset and graph construction

ClinVar variant annotations were obtained from the official NCBI ClinVar FTP tab-delimited variant-summary file [13]. We used the ClinVar variant-summary release dated 31 March 2024 and retained checksums for the exact input file used in the reconstruction. We reconstructed a filtered, non-conflicting missense dataset and mapped variants to STRING human protein identifiers. We retained GRCh38 germline missense SNVs with criteria-supported review status, combined pathogenic and likely pathogenic records into the positive class, combined benign and likely benign records into the negative class, removed uncertain or conflicting records, and collapsed duplicate VariationIDs. The final ClinVar-STRING dataset contained 177,213 missense variants mapped to 14,123 labelled STRING proteins. The propagation graph contained 19,622 STRING protein nodes and 13,715,404 directed edge-index entries derived from the STRING v12 protein-association network, which was used as a PPI-context graph rather than as a pure physical-interaction map [24].

B. Protein representations and PPI propagation

Raw protein features were generated using the ESM-2 650M checkpoint, with 1,280-dimensional protein-level vectors obtained by residue averaging [14]. PPI-GCN embeddings were computed by fixed two-hop GCN-style propagation over the normalized STRING graph [11]. We use "PPI-GCN" as shorthand for fixed GCN-style propagation of ESM features; no graph-convolution parameters were learned. This isolates topology-induced smoothing from capacity gains due to trainable graph layers, yielding a controlled propagated representation rather than an end-to-end GNN predictor. The implementation symmetrized the retained edge list, added self-loops, coalesced duplicate edges, used symmetric degree normalization and averaged the raw, one-hop and two-hop features. Propagation did not use labels; downstream classification used the representation of each variant's mapped STRING protein. Because propagation used the full graph containing train, validation and test nodes and their ESM features, the setting is a transductive shared-graph audit rather than a fully inductive predictor.

C. Splits, classifiers and external predictors

The original split assigned train, validation and test variants by STRING protein ID, with no shared proteins across partitions. The split contained 141,864 training variants, 17,129 validation variants and 18,220 test variants. The homology-aware split used MMseqs-derived protein clusters so that homologous clusters were assigned wholly to one partition while preserving the same row counts [22]. This split is treated as a stress test rather than a prevalence-matched replacement because the test positive rate changed from 31.95% in the original split to 10.56% in the homology-aware split.

All representation and fusion models were evaluated with a downstream MLP classifier selected by validation AUC and optimized with Adam [10]. Multi-feature representation and fusion models used a 512/128 hidden-layer MLP with dropout 0.3. Low-dimensional external-score baselines used a smaller 32/16 classifier with dropout 0.2, while architecture-matched CADD+AlphaMissense and fusion-control comparisons used the 512/128 classifier to keep downstream capacity matched. Training used mini-batches of 1,024 variants, learning rate 10^{-3}, weight decay 10^{-5}, a maximum of 50 epochs and early stopping after five consecutive validation checks without improvement in validation AUC. Class imbalance was handled with `BCEWithLogitsLoss` using training-partition positive-class weight $N_{\text{negative}}/N_{\text{positive}}$. CADD GRCh38 v1.7 scores and AlphaMissense hg38 scores were matched to the reconstructed variants [12], [21], [5]. CADD scores covered 177,205 variants (99.995%), and AlphaMissense scores covered 171,753 variants (96.92%). Missing CADD and AlphaMissense scores were imputed with training-set medians; AlphaMissense-based feature sets also included a binary missingness indicator. In the evaluated MLP configurations, no additional z-score standardization was applied to CADD, AlphaMissense or embedding-derived inputs. The architecture-matched fusion controls share classifier capacity and feature construction but do not test alternative feature-scaling choices. Repeated classifier comparisons used seeds 42, 43 and 44.

D. Audit analyses

Uncertainty was assessed using both variant-level bootstrap and protein-clustered bootstrap [6]. To test connectivity dependence, we removed direct train-test PPI edges and regenerated PPI-GCN embeddings. Partial edge-ablation curves removed 0%, 25%, 50%, 75% and 100% of direct train-test protein pairs using random removal, higher-STRING-score-first removal and lower-STRING-score-first removal. To test whether generic graph smoothing alone could explain the gain, topology permutation preserved the unlabeled graph topology while randomizing the correspondence between protein identity and graph position. Follow-up fusion controls also concatenated CADD+AlphaMissense features with raw ESM, edge-ablated PPI-GCN and topology-permuted PPI-GCN representations to distinguish protein-representation complementarity from graph-topology-specific complementarity.

Topology-context analyses stratified test variants by degree, k-core, betweenness, clustering coefficient, neighbour-risk and community-boundary features. Neighbour-risk and neighbour pathogenic-rate features for validation and test proteins were computed from training-set labels only; validation and test labels were not used to construct these topology-context features. Corrected-sample analyses compared variants whose predictions were corrected or degraded after adding graph context. Multiple topology-feature comparisons used Benjamini-Hochberg false-discovery-rate correction [2]. AUPRC was interpreted within each split because precision-recall metrics are sensitive to class prevalence [20].

III. Results

A. Full-graph PPI-GCN improves point estimates but depends on graph connectivity

Under the original protein-level split, a retained reference run used for paired graph-ablation and bootstrap analyses gave raw ESM test AUC 0.7790 and AUPRC 0.6074. PPI-GCN improved these values to AUC 0.7997 and AUPRC 0.6496, corresponding to Δ AUC = +0.0207 and Δ AUPRC = +0.0421 (Fig. 1B). Across three repeated classifier seeds, the comparison remained directionally positive but with a smaller mean AUC difference: raw ESM reached AUC 0.7835 ± 0.0051 and AUPRC 0.6116 ± 0.0054, whereas PPI-GCN reached AUC 0.7950 ± 0.0013 and AUPRC 0.6484 ± 0.0096. Variant-level bootstrap of the retained reference run gave Δ AUC +0.0208, 95% CI [0.0174, 0.0240], and Δ AUPRC +0.0420, 95% CI [0.0353, 0.0484]. Protein-clustered bootstrap widened both intervals across zero: Δ AUC CI [-0.0080, +0.0527] and Δ AUPRC CI [-0.0021, +0.0798] (Fig. 1C).

The full graph contained 1,242,422 direct train-test edge-index entries, representing 9.06% of all edge-index entries and 621,211 unique undirected train-test protein pairs. Removing all direct train-test PPI edges reduced edge-ablated ESM-GCN to AUC 0.7791 ± 0.0076 and AUPRC 0.6255 ± 0.0148 (Fig. 1D). Relative to raw ESM, the AUC gain was nearly eliminated (Δ AUC +0.0001), while a smaller AUPRC gain remained (Δ AUPRC +0.0181). Partial ablation further showed that AUC gain declined as direct train-test pairs were removed, with higher-STRING-score edge removal generally causing stronger loss at intermediate and high removal fractions (Fig. 1E).

When graph topology was preserved but protein identities were randomly reassigned to graph positions, topology-permuted ESM-GCN achieved AUC 0.7757 ± 0.0067 and AUPRC 0.6052 ± 0.0086, close to raw ESM and below real PPI-GCN (Fig. 1F). The retained full-graph gain was consistent with protein-specific topology alignment rather than generic graph smoothing, while its AUC component remained sensitive to cross-split connectivity.

B. PPI context is complementary under the original split but bounded by strong predictors and homology-aware evaluation

Under the original split, the external variant-effect predictors provided strong baselines. The single-score CADD and AlphaMissense baselines achieved CADD-only AUC 0.9207 and AUPRC 0.8049, and AlphaMissense-only AUC 0.9470 and AUPRC 0.8899. In the matched repeated-run fusion-control audit used for Fig. 2, an architecture-matched CADD+AlphaMissense baseline trained with the same 512/128 MLP used for fusion controls achieved mean AUC 0.9572 and AUPRC 0.9068. Adding raw ESM representation to CADD +AlphaMissense improved performance to AUC 0.9680 and AUPRC 0.9317, showing complementarity between protein-level sequence representation and the two scalar predictors. Real PPI-GCN fusion achieved AUC 0.9718 and AUPRC 0.9415; edge-ablated graph fusion reached AUC 0.9708 and AUPRC 0.9395, and topology-permuted graph fusion reached AUC 0.9694 and AUPRC 0.9355. Paired bootstrap using ensemble-averaged predictions gave a real-fusion margin over raw ESM fusion (protein-clustered Δ AUC 95% CI 0.0005 to 0.0075; Δ AUPRC CI 0.0032 to 0.0167). The smaller margins over edge-ablated and topology-permuted graph controls crossed zero under protein-clustered uncertainty. Most of the original-split fusion gain was retained by protein-representation controls, with a small topology-specific margin for real PPI-GCN (Fig. 2B,C).

Under the homology-aware split, PPI-GCN remained higher than raw ESM, increasing AUC from 0.7248 to 0.7606 and AUPRC from 0.2418 to 0.2768 (Fig. 2D). With CADD and AlphaMissense included, the retained graph-fusion estimate did not exceed the repeated-run architecture-matched two-predictor baseline. The architecture-matched CADD+ AlphaMissense baseline achieved AUC 0.9754 and AUPRC 0.8401; CADD+AlphaMissense plus raw ESM achieved mean AUC 0.9728 and AUPRC 0.8249; and the retained graph-fusion model achieved AUC 0.9741 and AUPRC 0.8333 (Fig. 2E). A retained-reference bootstrap based on a different comparison protocol also did not show a positive margin. PPI context remained useful relative to raw ESM, but not beyond the strong predictors in this stress test.

C. Graph gains concentrate in specific topology contexts

ESM representation space showed weak but detectable correspondence with PPI topology. Sampled PPI-neighbour pairs had mean ESM cosine similarity 0.8789 compared with 0.8639 for random protein pairs, and within-community pairs had higher similarity than between-community pairs (0.8743 versus 0.8642; Fig. 3A). This pattern is consistent with modest correspondence between ESM representation space and PPI network structure before propagation.

Graph-derived gains were not monotonic with simple centrality. Mid-k-core variants showed Δ AUC +0.0775 and Δ AUPRC +0.1049, whereas high-degree variants showed Δ AUC -0.0178 (Fig. 3B). Betweenness and clustering also shaped gain profiles: low- and mid-betweenness groups showed positive AUC gains (+0.0614 and +0.0522), while high betweenness showed a negative AUC difference (-0.0145); mid- and high-clustering groups showed positive gains, while low clustering was close to zero (Fig. 3C).

Neighbour-risk stratification and joint topology maps showed that the graph signal was concentrated in local contexts rather than uniformly distributed. Low-, mid- and high-neighbour-risk strata all had positive AUC gains (+0.0641, +0.0265 and +0.0284), but the shape of AUC and AUPRC gains differed across strata (Fig. 3D). Exploratory joint topology maps further identified stronger positive and negative profiles, including mid-k-core plus high-betweenness variants (Δ AUC +0.1429), mid-k-core plus mid-clustering variants (+0.1408), high-neighbour-risk plus mid-k-core variants (+0.1274), and high-k-core plus low-clustering variants (-0.0532; Fig. 3E,F). These cells are descriptive predictive gain profiles selected within this dataset, not independently validated biological subtypes; their bootstrap intervals quantify

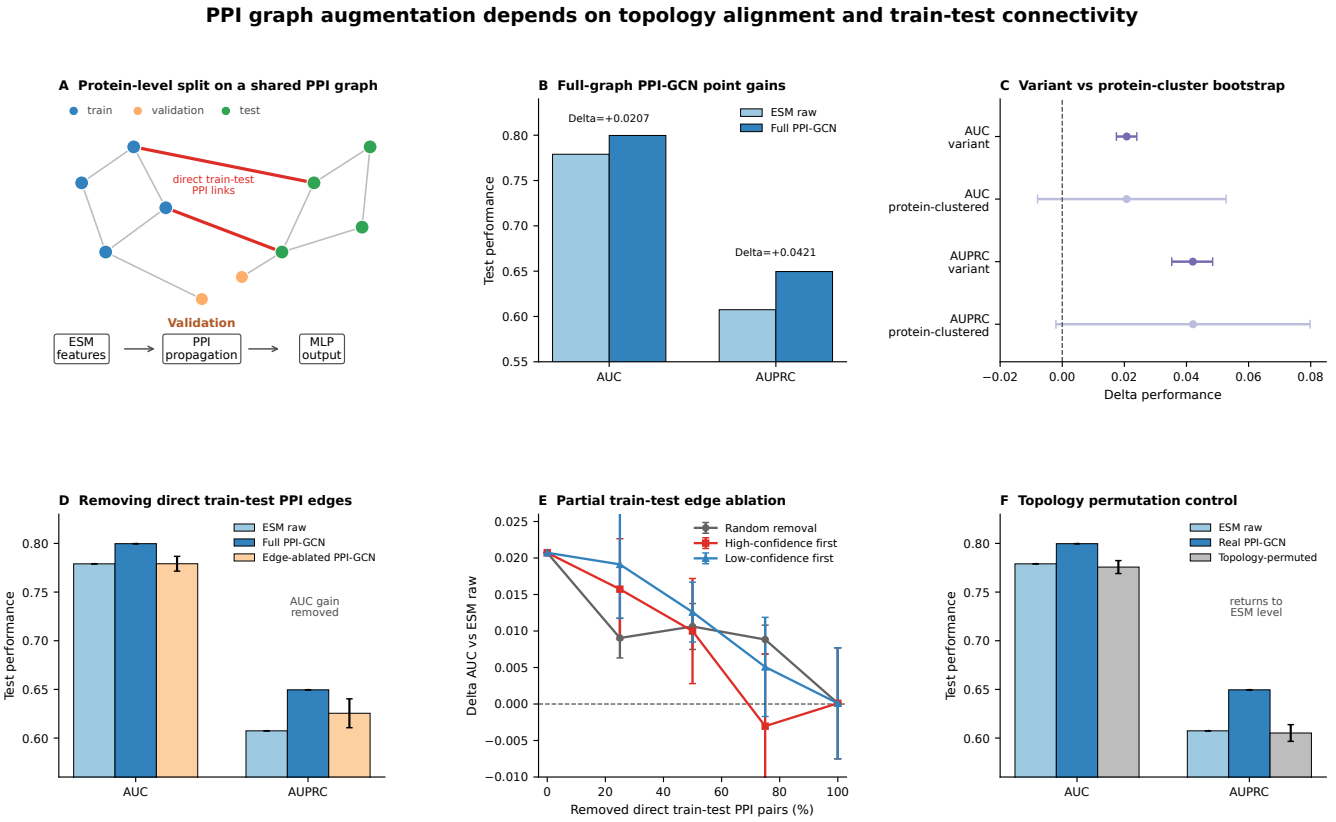


Fig. 1. PPI graph augmentation is sensitive to protein-topology correspondence and train-test connectivity. A, Original protein-level split on a shared PPI graph. B, Retained reference run in which full-graph PPI-GCN improves over raw ESM; repeated classifier seeds show the same direction with a smaller mean AUC difference, as reported in the text and Supplementary Table 2. C, Variant-level bootstrap supports positive gain, whereas protein-clustered bootstrap widens uncertainty across zero. D, Removing direct train-test PPI edges nearly eliminates the reference-run AUC gain. E, Partial train-test edge ablation shows connectivity-sensitive degradation. F, Topology permutation returns performance close to raw ESM.

within-cell sampling uncertainty rather than selection-adjusted reproducibility.

D. Graph augmentation reshapes false-negative and false-positive predictions differently

At a fixed decision-score threshold of 0.5, raw ESM and PPI-GCN were jointly correct for 11,875 test variants and jointly wrong for 3,965 variants. PPI-GCN corrected 1,436 variants and degraded 944 variants relative to raw ESM. Corrected variants included 837 pathogenic false negatives rescued and 599 benign false positives suppressed, whereas degraded variants included 904 benign true negatives converted to false positives and 40 pathogenic true positives converted to false negatives (Fig. 4A).

As an exploratory retained-reference analysis, the transition from the earlier CADD+AlphaMissense baseline to the corresponding retained graph-fusion model had a different structure from the raw-ESM transition. The two retained models were jointly correct for 15,981 variants and jointly wrong for 1,298 variants; graph fusion corrected 666 variants and degraded 275 variants. The corrected set contained more suppressed false positives than rescued false negatives (444 versus 222; Fig. 4B). This transition analysis was not recomputed from the architecture-matched repeated-run ensembles used in Fig. 2.

Topology profiles differed between rescued false negatives and suppressed false positives. In the raw ESM to PPI-GCN comparison, rescued false negatives had higher degree, neighbour-risk, neighbour pathogenic rate, neighbour training-variant count and betweenness, whereas suppressed false positives had higher clustering coefficient and within-community neighbour fraction (Fig. 4C). Similar but not identical profile shifts appeared in the retained CADD+AlphaMissense to graph-fusion comparison (Fig. 4D). Multiple topology features remained significant after FDR correction (Fig. 4E), and difficulty-matched contrasts retained differences across degree, k-core, neighbour-risk, neighbour pathogenic rate, betweenness and boundary role (Fig. 4F). Graph-related prediction changes were associated with network context; the analysis was descriptive rather than causal.

IV. Discussion

PPI graph augmentation improved benchmark performance in the shared-graph evaluation, with effects that varied by evaluation design and baseline choice. PPI-GCN improved over raw ESM in the original split, and the gain was sen-

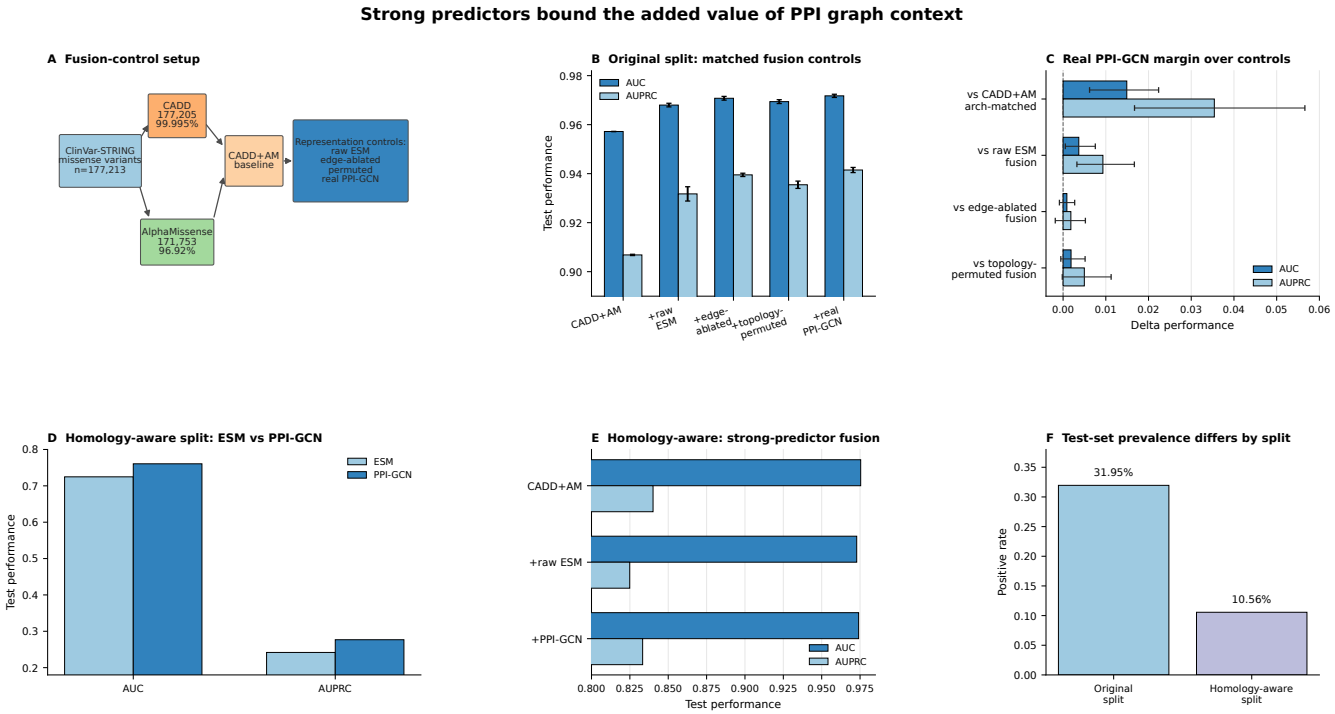


Fig. 2. PPI graph context is bounded by protein-representation controls and homology-aware evaluation. A, CADD and AlphaMissense matching and fusion-control setup. B, Original-split repeated-run performance of architecture-matched CADD+AlphaMissense, CADD+AlphaMissense+raw ESM, CADD +AlphaMissense+edge-ablated PPI-GCN, CADD+AlphaMissense+topology-permuted PPI-GCN and CADD+AlphaMissense+real PPI-GCN. Bars show mean test performance across three matched classifier seeds except topology-permuted fusion, which uses three topology permutations by three classifier seeds (n=9); error bars indicate one standard deviation. C, Marginal test-set gain of real PPI-GCN fusion over architecture-matched CADD+AlphaMissense and protein-representation controls; error bars show protein-clustered paired-bootstrap 95% confidence intervals computed from ensemble-averaged test predictions. Because panel C computes metrics from ensemble-averaged probabilities, its point deltas need not equal differences between the across-seed mean metrics shown in panel B. The topology-permuted interval conditions on the sampled topology permutations and does not include permutation-generation uncertainty. D, Homology-aware split comparison of raw ESM and PPI-GCN. E, Homology-aware architecture-matched CADD+AlphaMissense, CADD+AlphaMissense+raw ESM and graph fusion; CADD+AlphaMissense and CADD+AlphaMissense+raw ESM are repeated-run means, whereas graph fusion is the retained homology-aware point estimate. F, Test-set prevalence differs between original and homology-aware splits.

sitive to protein-topology correspondence and cross-split connectivity. With CADD+AlphaMissense included, raw ESM, edge-ablated PPI-GCN and topology-permuted PPI-GCN all produced large fusion gains, while real PPI-GCN was only modestly higher. The strong-predictor fusion gain primarily reflects protein-representation complementarity, with a smaller real-topology margin.

PPI-GCN remained useful relative to raw ESM under the homology-aware split, but the retained graph-fusion estimate did not exceed the repeated-run architecture-matched CADD +AlphaMissense baseline when homologous protein clusters were kept within partitions. The split also changed test-set prevalence and serves as a stress test rather than a prevalence-matched replacement. Claims about graph complementarity depend on the baseline predictor, split design and graph-control baseline.

Graph propagation changed predictions in structured topology contexts. The observed patterns may reflect differences between proteins embedded in densely connected functional neighbourhoods and those occupying more boundary-like network positions. They do not establish a specific molecular mechanism and require replication in independent cohorts, temporal ClinVar splits, disease-specific networks and alternative PPI sources.

ClinVar labels may reflect submission bias and future label updates [13]. Historical overlap between the benchmark and variants used during development or evaluation of CADD and AlphaMissense was not audited, so their scores represent benchmark rather than prospective performance. Variant mapping was performed at the protein-identifier level without separate modeling of isoform-specific effects or transcript ambiguity. STRING combined scores integrate multiple evidence channels and do not represent direct physical interaction strength [24]. The original split is transductive; direct-edge ablation leaves train-validation, validation-test and higher-order paths intact. Validation AUC was used for checkpoint selection, and the retained PPI-GCN run showed a larger validation gain than test gain. The conclusions rely on held-out test metrics, repeated-seed summaries and protein-clustered uncertainty. The homology-aware split changes class prevalence, limiting direct AUPRC comparisons across splits. Figure 4 transition counts depend on a fixed decision-score threshold, and class-weighted sigmoid outputs are not calibrated clinical probabilities. Alternative scaling of scalar and high-

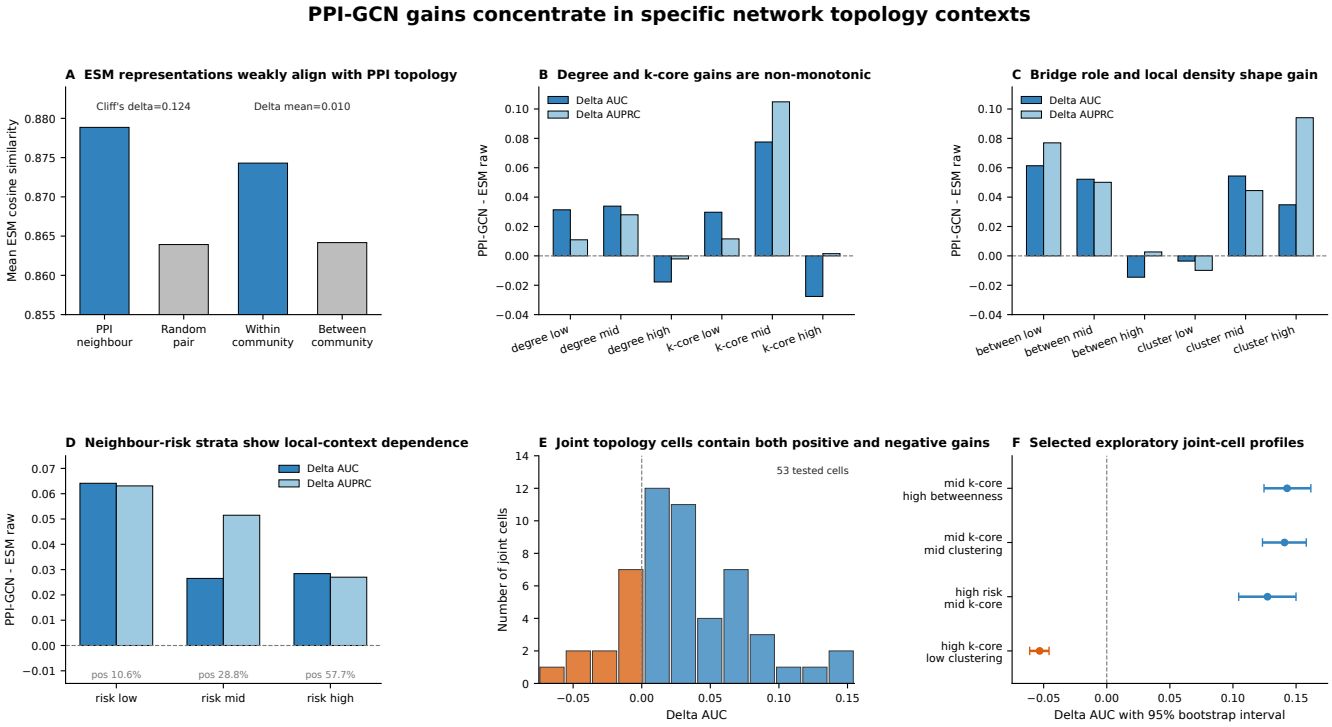


Fig. 3. PPI-GCN gains concentrate in specific network topology contexts. A, ESM representations show weak alignment with PPI topology. B, Degree and k-core gains are non-monotonic. C, Betweenness and clustering strata shape graph-derived gain. D, Neighbour-risk strata show local-context dependence. E, Joint topology cells contain both positive and negative gains. F, Selected exploratory joint-cell profiles illustrate heterogeneous gains. The joint-cell panels are descriptive and were not used for cell-level multiple-testing inference.



Fig. 4. Graph augmentation changes false-negative and false-positive predictions in different network contexts. A, Prediction transitions from raw ESM to PPI-GCN at a fixed decision-score threshold of 0.5. B, Retained-reference CADD+AlphaMissense to graph-fusion transitions; this exploratory transition analysis was not recomputed from the architecture-matched repeated-run ensembles in Fig. 2. C-D, Row-normalized heatmaps compare topology profiles of suppressed false positives and rescued false negatives. E, Corrected groups differ across multiple topology features after FDR correction. F, Difficulty-matched contrasts retain topology differences.

dimensional inputs was not exhaustively evaluated; identical preprocessing controls relative comparisons, but absolute fusion performance may depend on these choices. The audit relies on retained processed outputs and a compact source-data archive, while provider-controlled resources such as CADD and AlphaMissense must be obtained from their original sources. PPI graph augmentation is best treated as a conditional source of network context rather than a substitute for strong sequence or variant-effect predictors.

Data and Code Availability

A compact anonymized source-data archive containing figure source data, summary metrics, bootstrap outputs and checksums has been prepared. The archive and analysis scripts will be released according to venue policy. Large third-party resources, including CADD, AlphaMissense, ClinVar and STRING source files, should be obtained from their original providers.

Ethical Statement and AI-Tool Disclosure

This study used publicly available aggregate variant annotations and computationally derived representations. It did not involve direct recruitment of human participants, intervention, or access to identifiable individual-level clinical records.

OpenAI ChatGPT, OpenAI Codex, and Anthropic Claude were used for language editing, literature organization, consistency checking, and assistance with manuscript drafting, figure-caption preparation, and analysis-script drafting and debugging. Study design, model comparisons, evaluation criteria, experiment execution, result interpretation, numerical verification, and citation checking remained under human control. All reported numerical results and citations were verified against retained analysis outputs, source-data files, and original references.

The AI tools did not act as authors, independently determine the scientific conclusions, or generate or fabricate experimental data. Full responsibility for the study and the final manuscript rests with the authors.

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